

2020/2021: Tutorial Sheet 4

2-dim nonlinear ODEs

I. Explicit limit cycle Find, using plane-polar coordinates or otherwise, the explicit limit cycle for the following non-linear systems of ODEs. Determine also if it is stable or unstable.

$$\begin{aligned}\frac{dx}{dt} &= 4x + y - x(x^2 + y^2), \\ \frac{dy}{dt} &= 4y - x - y(x^2 + y^2).\end{aligned}$$

II. Poincaré-Bendixon Theorem

Consider the ODE system

$$\begin{aligned}\frac{dx}{dt} &= 2x + 2y - x(2x^2 + y^2), \\ \frac{dy}{dt} &= -2x + y - y(x^2 + 2y^2).\end{aligned}$$

1. You may assume that the only equilibrium point in the system is $(0, 0)$. Show, by considering linearisation and asymptotic arguments, that there must be a limit-cycle solution occurring.
2. By considering solution-path directions across the circles

$$x^2 + y^2 = \text{constant},$$

find an annular region which must contain the limit cycle solution.

III. Hopf bifurcations

Each of the following two-dimensional systems of non-linear ODEs possesses a limit-cycle solution for certain values of the parameter a . Assume $(0, 0)$ is the only equilibrium point and investigate the nature of the Hopf bifurcation that occurs at the critical value of a and state what the critical value is in each case.

(a)

$$\begin{aligned}\frac{dx}{dt} &= ax - y + x \left(x^2 + \frac{3}{2}y^2 \right) \\ \frac{dy}{dt} &= x + ay + y \left(x^2 + \frac{3}{2}y^2 \right)\end{aligned}$$

(b)

$$\begin{aligned}\frac{dx}{dt} &= -ax - y + x \left(x^2 + 3y^2 \right) \\ \frac{dy}{dt} &= x - ay + y \left(x^2 + 3y^2 \right)\end{aligned}$$